

HYDROGENATED NBR SYNTHESIS AND LOW-MOLECULAR-WEIGHT HNBR PROCESS PATENTS

1. ABSTRACT

This report synthesizes five primary patents focused on hydrogenated nitrile rubber (HNBR) preparation, process intensification, and post-reaction purification. The target material is hydrogenated NBR derived from acrylonitrile-butadiene rubber, with emphasis on low residual unsaturation, improved heat/oil resistance, and better processability through control of molecular weight, Mooney viscosity, and catalyst residues. Across the selected patents, two major synthesis strategies dominate: direct selective hydrogenation of NBR using Ru/Os-based catalyst systems, and single-step or tandem metathesis/hydrogenation routes that reduce molecular weight while hydrogenating the polymer backbone. A third theme is downstream purification, especially removal of ruthenium residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins.

US10500578B2 and EP2473281B1 are the most process-centric disclosures for low-molecular-weight HNBR. They describe catalyst compositions based on ruthenium or osmium complexes, often paired with vinyl ether co-catalysts, enabling simultaneous metathesis and hydrogenation or metathesis followed by hydrogenation. These routes are designed to produce HNBR with reduced Mw, narrower PDI, and low residual double bond content. US10500578B2 provides detailed comparative examples showing the effect of catalyst pretreatment with vinyl ethyl ether on hydrogenation rate and molecular-weight retention. EP2473281B1 similarly emphasizes one-step production of HNBR with Mw 20,000-250,000 and Mooney viscosity 1-50.

US7772328B2 is distinct in that it focuses on low-molecular-weight HNBR made by olefin metathesis of an already hydrogenated nitrile rubber framework, targeting low Mooney viscosity and narrow molecular-weight distribution rather than the hydrogenation catalyst itself.

US9150669B2 reinforces the same single-step simultaneous metathesis/hydrogenation concept but highlights alternative catalyst structures and performance benchmarks. US7846995B2 addresses a practical manufacturing issue: removal of ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber using macroreticular functionalized ion-exchange resins. Together, the patents show a coherent development path from catalyst design and reaction control to purification and product stabilization for industrial HNBR manufacture.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight hydrogenated nitrile rubber prepared by olefin metathesis of nitrile rubber using a ruthenium-based metathesis catalyst and a co-olefin, followed by hydrogenation to obtain HNBR. The process is aimed at reducing molecular weight and Mooney viscosity while preserving the nitrile rubber microstructure. - Cement concentration: 15 wt.-% solid — Example 1 - Rubber mass: 200 g — Example 1 - MCB mass: 1133 g — Example 1 - Degassing cycles: 3 times with C₂H₄ — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber mass: 75 g — Example 3 - MCB mass: 1175 g — Example 3 - Cement concentration: 6 wt.-% solid — Example 3 - 1-hexene mass: 150 g — Example 4 - Co-olefin pressure: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - PDI: 2.0 — typical product - Mn: 85 kg/mol — starting polymer - Mw: 296 kg/mol — starting polymer - PDI: 3.4 — starting polymer - Mw: 189000 — Experiment 1 - Mw: 136000 — Experiment 2 - Mw: 59000 — Experiment 3 - Mw: 111000 — Experiment 4 - PDI: 2.59 — Experiment 1 - PDI: 2.27 — Experiment 2 - PDI: 1.90 — Experiment 3 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 43 — Experiment 1 - Mooney viscosity: 28 — Experiment 2 - Mooney viscosity: 3 — Experiment 3 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis conditions were used on a 15 wt.-% rubber cement with 6.9×10^5 Pa ethylene. - Example 2: A second ethylene metathesis run was performed under the same degassing and cement conditions. - Example 3: Lower-solids cement (6 wt.-%) was processed under ethylene to further reduce molecular weight. - Example 4: 1-hexene was used as co-olefin at 80° C., showing the effect of co-olefin choice and catalyst loading on product properties.

Technical Relevance This patent is directly relevant because it teaches a route to low-molecular-weight HNBR through metathesis-based chain scission followed by hydrogenation. It provides practical evidence that co-olefin selection, catalyst loading, and temperature can be used to tune Mw and Mooney viscosity.

Patent 2

- Patent Number: US10500578B2
- Patent Title: Catalyst compositions and their use for hydrogenation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses selective hydrogenation of nitrile rubber using Ru- or Os-based catalyst compositions that are precontacted with a vinyl-containing co-catalyst, especially ethyl vinyl ether. The examples show that catalyst pretreatment can accelerate hydrogenation and influence the molecular-weight profile of the resulting HNBR, and the disclosure also includes a metathesis-plus-hydrogenation sequence. - NBR concentration: 6 wt % — Example 1 - Rubber mass: 18 g — Example 1 - MCB mass: 282 g — Example 1 - Nitrogen bubbling time: 30 minutes — Example 1 - Reaction temperature: 120° C. — Example 1 - Wilkinson's catalyst: 15 mg — Example 1 - PPh₃: 18 mg — Example 1 - Hydrogen pressure: 4.137 MPa — Example 1 - Agitation speed: 800 rpm — Example 1 - Hydrogenation time: 5 hours — Example 1 - Hydrogenation degree: 90.3% — Example 1 - Catalyst (2): 9 mg — Example 2 - NBR concentration: 3 wt % — Example 2 - Hydrogenation time: 4 hours — Example 2 - Hydrogenation degree: 98.6% — Example 2 - Catalyst (2): 18 mg — Example 3 - Hydrogenation time: 18 hours — Example 3 - Hydrogenation degree: 92.6% — Example 3 - Vinyl ethyl ether: 12 hours pretreatment — Example 4 - Hydrogenation time: 2 hours — Example 4 - Hydrogenation degree: 99% — Example 4 - Vinyl ethyl ether: 100 µL — Example 5 - Hydrogenation time: 4 hours — Example 5 - Hydrogenation degree: 95% — Example 5 - Hydrogenation degree: 96% — Example 6 - Hydrogenation degree: 99% — Example 7 - Hydrogen pressure: 6.895 MPa — Example 8 - Hydrogenation time: 3 hours — Example 8 - Hydrogenation degree: 98% — Example 8 - Catalyst (1): 9 mg — Example 9 - Hydrogenation time: 3 hours — Example 9 - Hydrogenation degree: 97% — Example 9 - Metathesis time: 1 hour — Example 10 - VEE addition: 0.5 mL — Example 10 - Hydrogenation time: 3 hours — Example 10 - Hydrogenation degree: 99% — Example 10 - Metathesis time: 2 hours — Example 11 - Hydrogenation time: 3 hours — Example 11 - Hydrogenation degree: 99% — Example 11 - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 0.4% — Table 2 - RDB: 3.5% — Table 2 - RDB: 0.7% — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Comparative hydrogenation with Wilkinson's catalyst reached 90.3% hydrogenation in 5 hours. - Example 2: Catalyst (2) without pretreatment

gave 98.6% hydrogenation in 4 hours at lower feed concentration. - Example 4: VEE pretreatment enabled 99% hydrogenation in only 2 hours. - Example 8: Higher hydrogen pressure (6.895 MPa) still gave 98% hydrogenation in 3 hours. - Example 10: A metathesis step followed by hydrogenation produced 99% hydrogenation with reduced Mw and PDI.

Technical Relevance This patent is highly relevant because it discloses catalyst compositions specifically designed for efficient HNBR hydrogenation and, in some embodiments, simultaneous metathesis/hydrogenation. It provides concrete process data showing how pretreatment and catalyst choice affect hydrogenation rate, RDB reduction, and molecular-weight retention.

Patent 3

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches a one-step process for producing low-molecular-weight hydrogenated nitrile rubber by reacting nitrile rubber in the presence of hydrogen and a ruthenium- or osmium-based metathesis catalyst, optionally with a co-olefin. The process is intended to combine chain scission and hydrogenation in a single operation to obtain HNBR with low residual double bond content and narrow molecular-weight distribution. - Mw: 20,000 to 250,000 — inventive process - Mooney viscosity: 1 to 50 — inventive process - PDI: less than 3.0 — inventive process - Hydrogenation degree: 50 to 100% — description - Hydrogenation degree: 80 to 100% — description - Hydrogenation degree: 90 to 99.9% — description - Remaining unsaturation: below 18% — commercial grades - Acrylonitrile content: up to about 50% — commercial grades - Mooney viscosity: 24 to 70 — nitrile rubbers used for hydrogenation/metathesis - Mooney viscosity: 28 to 40 — preferred nitrile rubbers used for hydrogenation/metathesis - Mw: 30,000 to 250,000 — cited prior art context - Mooney viscosity: 3 to 50 — cited prior art context - PDI: less than 2.5 — cited prior art context

Relevant Experimental Evidence - Example 1: Catalyst comparison showed Hoveyda and Arlt catalysts outperforming Wilkinson's catalyst in hydrogenation rate. - Example 1: HNBR 1 reached 0.4% RDB after 240 min and had Mw 113,371 g/mol and Mooney viscosity 33.3. - Example 1: HNBR 2 and HNBR 3 demonstrated the effect of catalyst choice on RDB, Mw, and Mooney viscosity. - Example 1: The data support simultaneous metathesis/hydrogenation as a route to low-molecular-weight HNBR.

Technical Relevance This patent is directly relevant because it defines the target product window for low-molecular-weight HNBR and discloses a simultaneous hydrogenation/metathesis route. It

is especially useful for identifying practical ranges for Mw, Mooney viscosity, and residual unsaturation in HNBR synthesis.

Patent 4

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight hydrogenated nitrile rubber by contacting nitrile rubber with a ruthenium- or osmium-based catalyst in the presence of hydrogen, optionally with a co-olefin, so that metathesis and hydrogenation occur simultaneously. The disclosure emphasizes catalyst structures that enable one-step reduction of molecular weight and residual unsaturation. - Mw: 20,000 to 250,000 — inventive process - Mooney viscosity: 1 to 50 — inventive process - PDI: less than 3.0 — inventive process - Hydrogenation degree: 50 to 100% — description - Hydrogenation degree: 80 to 100% — description - Hydrogenation degree: 90 to 99.9% — description - Remaining unsaturation: below 18% — commercial grades - Acrylonitrile content: up to about 50% — commercial grades - Mooney viscosity: 24 to 70 — nitrile rubbers used for hydrogenation/metathesis - Mooney viscosity: 28 to 40 — preferred nitrile rubbers used for hydrogenation/metathesis - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 0.4% — Table 2 - RDB: 3.5% — Table 2 - RDB: 0.7% — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Example 1: Hoveyda catalyst gave the fastest hydrogenation profile among the three catalysts compared. - Example 1: HNBR 1 reached 0.4% RDB after 240 min and had the lowest Mooney viscosity. - Example 1: Catalyst choice strongly affected Mw and PDI after hydrogenation. - Example 1: The data support a single-step route to low-molecular-weight HNBR with controlled residual unsaturation.

Technical Relevance This patent is relevant because it provides a concrete catalyst-composition framework for simultaneous hydrogenation and metathesis of NBR. It also supplies benchmark product-property targets and comparative data that are useful for optimizing HNBR synthesis

conditions.

Patent 5

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses post-treatment of optionally hydrogenated nitrile rubber solutions with functionalized macroreticular ion-exchange resins to remove ruthenium-containing catalyst residues. The process is a purification step rather than a synthesis route, but it is important for producing cleaner HNBR after metathesis/hydrogenation catalysis. - Resin mass: 0.5 g — Example 2A - Rubber solution mass: 180 g — Example 2A - Reaction temperature: ca. 100° C. — Example 2A - Reaction time: 66 hours — Example 2A - Drying temperature: 60° C. — Example 2A - Ru content after treatment: 10 ppm — Example 2A - Ru removal: 33% — Example 2A - Initial Ru content: 15 ppm — Table 2

Relevant Experimental Evidence - Example 2A: Lewatit® MonoPlus MP 500 reduced Ru content in nitrile rubber solution after 66 hours at about 100° C. - Example 2A: The treated rubber contained 10 ppm Ru, corresponding to 33% removal. - Example 2B: Untreated control rubber was used to establish the initial Ru baseline of 15 ppm.

Technical Relevance This patent is relevant because catalyst residue removal is a key downstream requirement for HNBR made by ruthenium-catalyzed hydrogenation or metathesis/hydrogenation. It addresses product purity and processability, which are critical for elastomer applications.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10500578B2 and US9150669B2 both focus on simultaneous hydrogenation/metathesis of nitrile rubber using Ru/Os catalyst systems, but US10500578B2 emphasizes catalyst pretreatment with vinyl ethyl ether and detailed hydrogenation profiles, while US9150669B2 emphasizes the general catalyst class and product-property windows.
- EP2473281B1 and US9150669B2 disclose the same core one-step low-molecular-weight HNBR concept, with overlapping target ranges for Mw (20,000-250,000), Mooney viscosity (1-50), and low PDI; EP2473281B1 additionally frames the process as a low-molecular-weight

HNBR preparation method with explicit catalyst comparison data.

- US7772328B2 differs from the simultaneous hydrogenation/metathesis patents by focusing on metathesis-driven molecular-weight reduction of nitrile rubber followed by hydrogenation, rather than a single-step hydrogenation/metathesis process.
- US7846995B2 is not a synthesis patent but a purification patent; it complements the synthesis disclosures by addressing ruthenium residue removal from optionally hydrogenated nitrile rubber solutions using ion-exchange resins.
- Across the process patents, co-olefin use is explicitly disclosed in US7772328B2, US10500578B2, EP2473281B1, and US9150669B2, with vinyl ethyl ether appearing as a key co-catalyst/co-olefin in the catalyst-pretreatment and metathesis-enabled routes.
- Temperature control is centered around 120° C. in US10500578B2 and around 80° C. in US7772328B2, while EP2473281B1 and US9150669B2 emphasize product-property targets more than a single fixed temperature in the extracted evidence.
- Hydrogen pressure is explicitly high in US10500578B2 (4.137 MPa, and 6.895 MPa in one example), whereas US7772328B2 highlights co-olefin pressure conditions (500 psi ethylene) during metathesis-based chain scission.

4. CONCLUSION

The evidence supports two practical synthesis routes to hydrogenated NBR: direct selective hydrogenation of NBR with Ru/Os catalysts, and metathesis-assisted molecular-weight reduction followed by or concurrent with hydrogenation. The strongest process guidance comes from US10500578B2, EP2473281B1, and US9150669B2, which collectively indicate that Ru/Os catalyst systems, vinyl ether pretreatment or co-olefin assistance, temperatures around 120° C., and hydrogen pressures in the 4-7 MPa range can deliver high hydrogenation degrees and low residual double bond content while maintaining manageable molecular weight. For low-Mw HNBR, the evidence favors catalyst systems that combine metathesis activity with hydrogenation activity and that can be tuned by co-catalyst pretreatment. For product cleanup, ruthenium-removal ion-exchange treatment is recommended to improve final purity. Overall, the best-supported synthesis target is HNBR with Mw roughly 20,000-250,000, Mooney viscosity below 50, and residual unsaturation below about 5% when high-performance applications are desired.

5. REFERENCES

- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- US10500578B2 | Catalyst compositions and their use for hydrogenation of nitrile rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10500578B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>

- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>